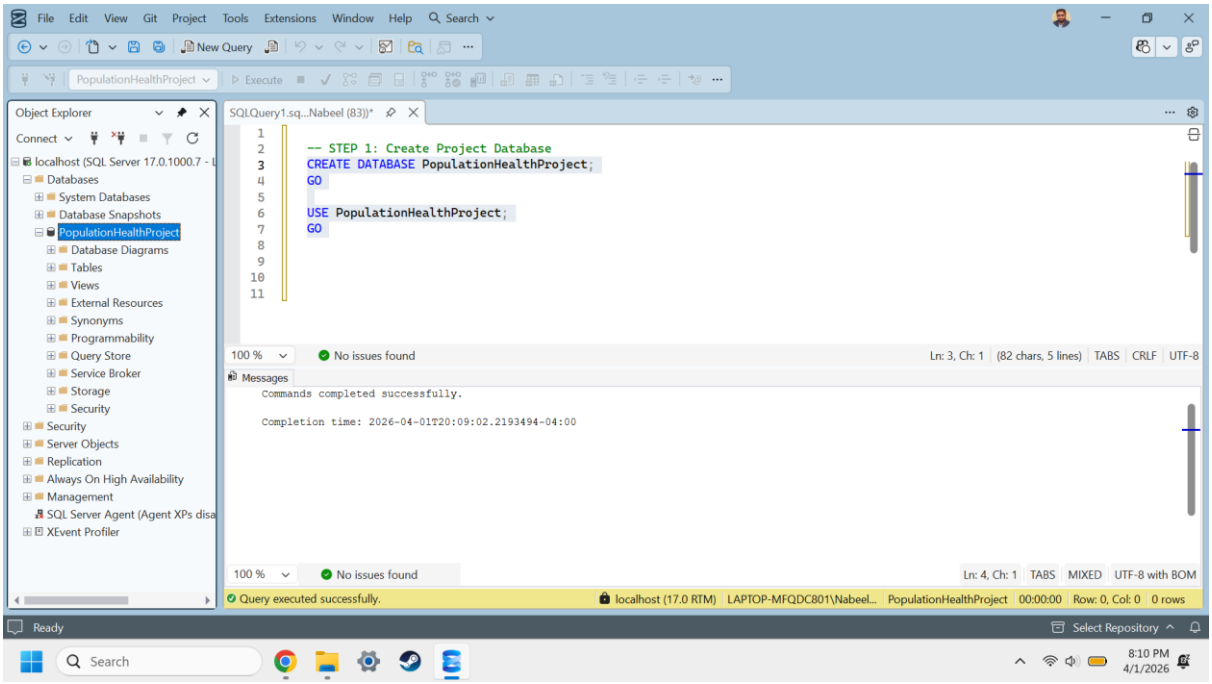
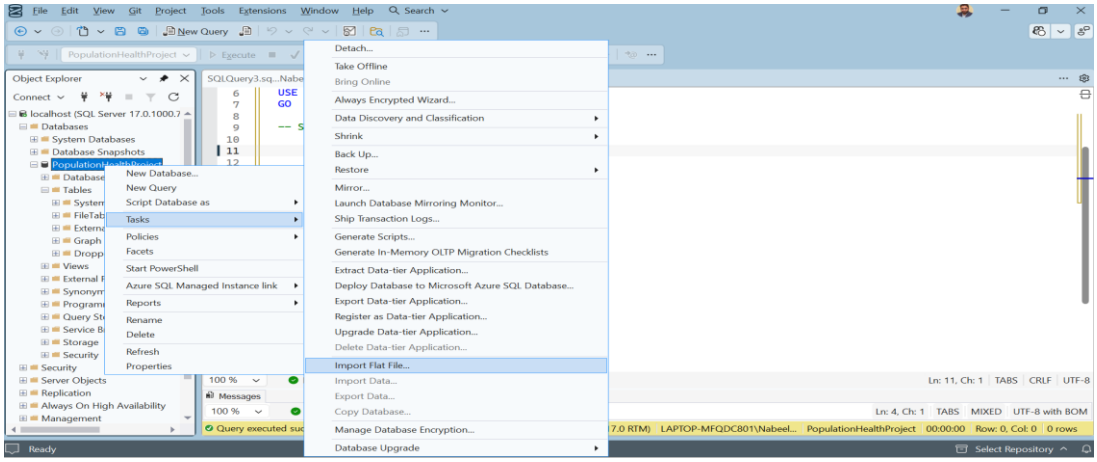
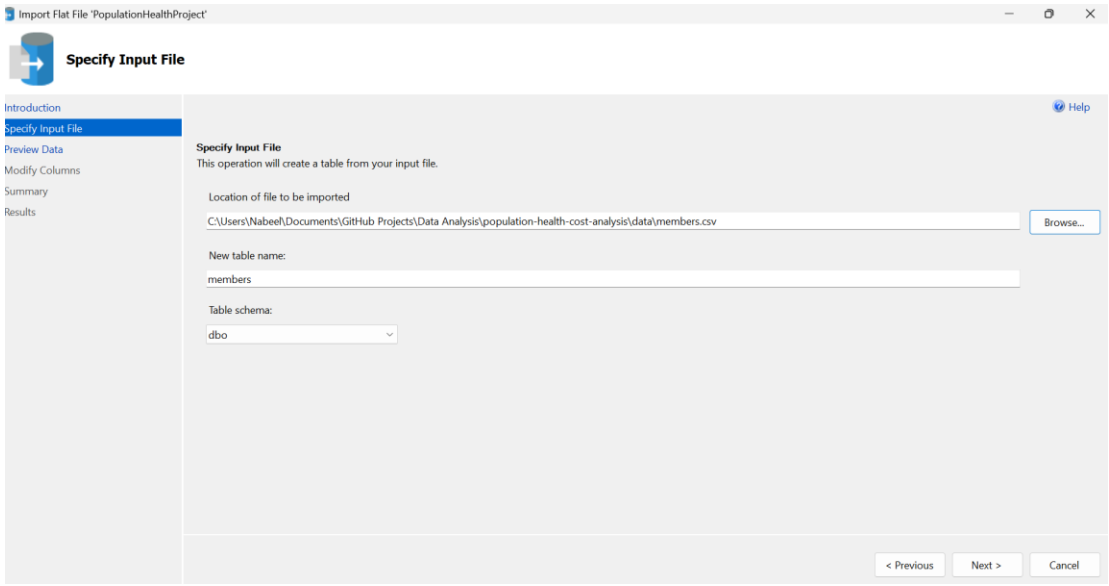
Create database

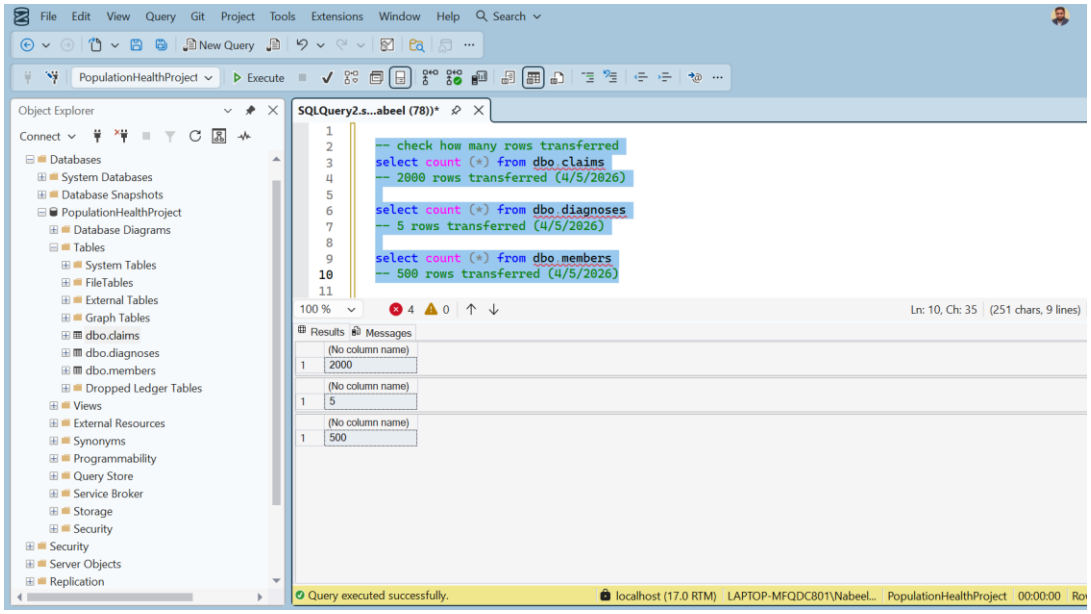


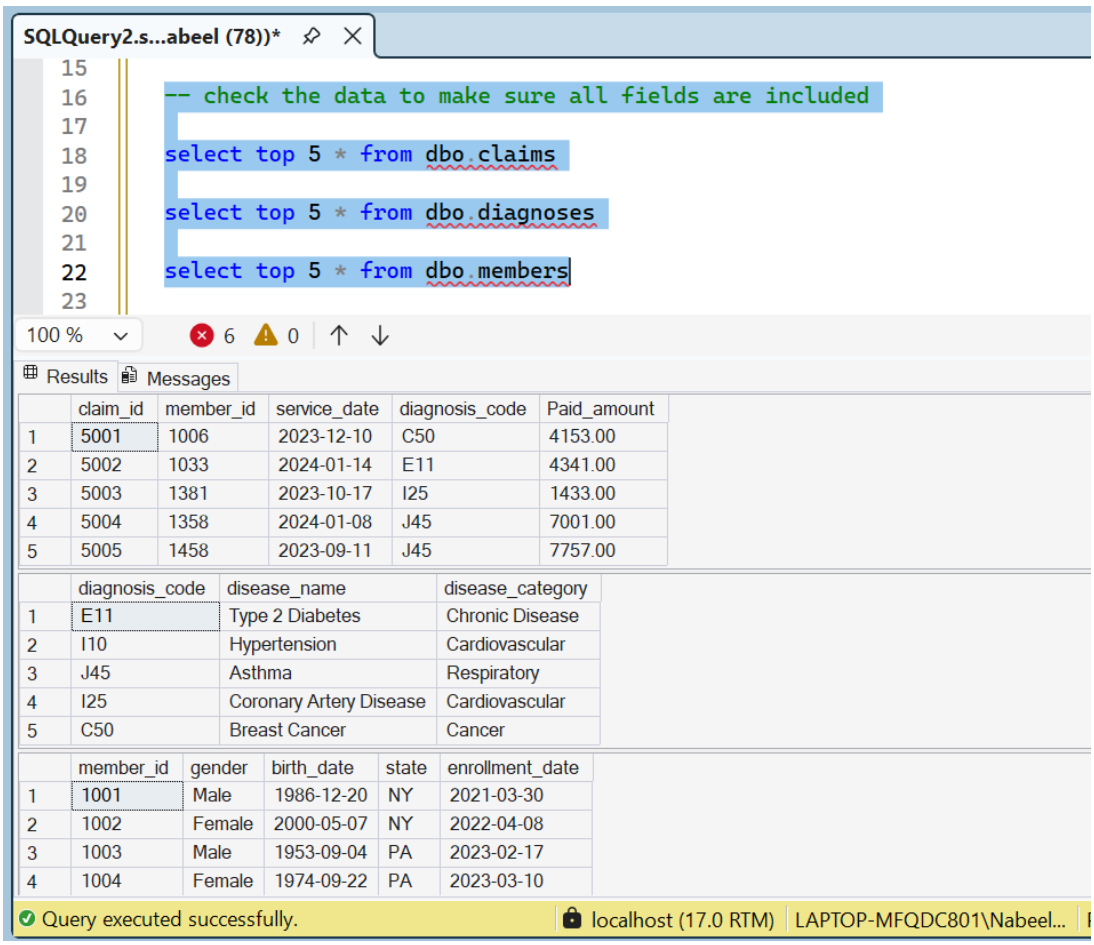
Import data from Excel (CSV fies) using "Import Flat File"





check if the data was imported correctly





	claim_id	member_id	service_date	diagnosis_code	Paid_amount
1	5001	1006	2023-12-10	C50	4153.00
2	5002	1033	2024-01-14	E11	4341.00
3	5003	1381	2023-10-17	I25	1433.00
4	5004	1358	2024-01-08	J45	7001.00
5	5005	1458	2023-09-11	J45	7757.00

	diagnosis_code	disease_name	disease_category
1	E11	Type 2 Diabetes	Chronic Disease
2	I10	Hypertension	Cardiovascular
3	J45	Asthma	Respiratory
4	I25	Coronary Artery Disease	Cardiovascular
5	C50	Breast Cancer	Cancer

	member_id	gender	birth_date	state	enrollment_date
1	1001	Male	1986-12-20	NY	2021-03-30
2	1002	Female	2000-05-07	NY	2022-04-08
3	1003	Male	1953-09-04	PA	2023-02-17
4	1004	Female	1974-09-22	PA	2023-03-10

disease prevalence analysis

SQLQuery2.s...abeel (78))*

```sql
-- disease prevalence analysis query

SELECT
d.disease_name,
COUNT(DISTINCT c.member_id) AS members_with_disease,
CAST(
ROUND(
COUNT(DISTINCT c.member_id) * 100.0 /
(SELECT COUNT(*) FROM members),2
) AS DECIMAL(5,2)
) AS prevalence_percent
FROM claims c
JOIN diagnoses d
ON c.diagnosis_code = d.diagnosis_code
GROUP BY d.disease_name
ORDER BY prevalence_percent DESC;
```

100 % 15 0

Results Messages

	disease_name	members_with_disease	prevalence_percent
1	Coronary Artery Disease	285	57.00
2	Type 2 Diabetes	285	57.00
3	Asthma	279	55.80
4	Hypertension	272	54.40
5	Breast Cancer	270	54.00

Query executed successfully. localhost (17.0 RTM) LAPTOP-MFQDC80

Financial Analysis

SQLQuery2.s...abeel (78))*

```sql
ORDER BY prevalence_percent DESC;

-- Financial Analysis
SELECT
d.disease_name,
COUNT(DISTINCT c.member_id) AS members,
SUM(c.paid_amount) AS total_cost,
AVG(c.paid_amount) AS avg_claim_cost
FROM claims c
JOIN diagnoses d
ON c.diagnosis_code = d.diagnosis_code
GROUP BY d.disease_name
ORDER BY total_cost DESC
```

100 % 24 0

Results Messages

	disease_name	members	total_cost	avg_claim_cost
1	Type 2 Diabetes	285	2111700.00	5137.9562
2	Hypertension	272	2006013.00	5027.6015
3	Asthma	279	2000303.00	4975.8781
4	Coronary Artery Disease	285	1909840.00	4715.6543
5	Breast Cancer	270	1813823.00	4735.8302

Query executed successfully. localhost (17.0 RTM) LAPTOP-MFQDC

Disease Prevelance by Age Group

```sql
-- Disease Prevelance by Age Group

-- Calculate age for each member

SELECT
member_id,
birth_date,
DATEDIFF(YEAR, birth_date, GETDATE()) AS age
FROM members
```

100 % 28 0

Results Messages

	member_id	birth_date	age
1	1001	1986-12-20	40
2	1002	2000-05-07	26
3	1003	1953-09-04	73
4	1004	1974-09-22	52
5	1005	1963-03-04	63
6	1006	1985-05-24	41
7	1007	1997-02-02	29
8	1008	1980-11-25	46
9	1009	2000-01-06	26
10	1010	1962-08-26	64
11	1011	1969-05-22	57

Query executed successfully. localhost (17.0 RTM) LA

SQLQuery2_1...abeel (78))*

```sql
-- Disease Prevelance by Age Group
-- Step 1 - Calculate Age for Each Member
SELECT
member_id,
birth_date,
DATEDIFF(YEAR, birth_date, GETDATE()) AS age
FROM members
```

100 % 99+ 0

Results | Messages

	member_id	birth_date	age
1	1001	1986-12-20	40
2	1002	2000-05-07	26
3	1003	1953-09-04	73
4	1004	1974-09-22	52
5	1005	1963-03-04	63
6	1006	1985-05-24	41
7	1007	1997-02-02	29
8	1008	1980-11-25	46
9	1009	2000-01-06	26
10	1010	1962-08-26	64
11	1011	1969-05-22	57
12	1012	1989-10-01	37
13	1013	1962-11-12	64
14	1014	2003-11-10	23
15	1015	1952-07-31	74
16	1016	1993-01-27	33

Query executed successfully. localhost (17.0 RTM) LAPTOP-MFQ

SQLQuery2_1...abeel (78))*

```sql
-- Step 2 - Assign Age Groups
SELECT
member_id,
birth_date,
DATEDIFF(YEAR, birth_date, GETDATE()) AS age,
CASE
    WHEN DATEDIFF(YEAR, birth_date, GETDATE()) BETWEEN 0 AND 17 THEN '0-17'
    WHEN DATEDIFF(YEAR, birth_date, GETDATE()) BETWEEN 18 AND 34 THEN '18-34'
    WHEN DATEDIFF(YEAR, birth_date, GETDATE()) BETWEEN 35 AND 49 THEN '35-49'
    WHEN DATEDIFF(YEAR, birth_date, GETDATE()) BETWEEN 50 AND 64 THEN '50-64'
    ELSE '65+'
END AS age_group
FROM members
```

100 % 99+ 0 Ln: 60, Cl

Results Messages

	member_id	birth_date	age	age_group
1	1001	1986-12-20	40	35-49
2	1002	2000-05-07	26	18-34
3	1003	1953-09-04	73	65+
4	1004	1974-09-22	52	50-64
5	1005	1963-03-04	63	50-64
6	1006	1985-05-24	41	35-49
7	1007	1997-02-02	29	18-34
8	1008	1980-11-25	46	35-49
9	1009	2000-01-06	26	18-34
10	1010	1962-08-26	64	50-64
11	1011	1969-05-22	57	50-64

Query executed successfully. localhost (17.0 RTM) LAPTOP-MFQDC801\Nabeel... PopulationHealth

SQLQuery2_1...abeel (78))*

```sql
--Step 3 – Join Claims with Members and Diagnoses

SELECT
c.member_id,
m.birth_date,
CASE
    WHEN DATEDIFF(YEAR, m.birth_date, GETDATE()) BETWEEN 0 AND 17 THEN '0-17'
    WHEN DATEDIFF(YEAR, m.birth_date, GETDATE()) BETWEEN 18 AND 34 THEN '18-34'
    WHEN DATEDIFF(YEAR, m.birth_date, GETDATE()) BETWEEN 35 AND 49 THEN '35-49'
    WHEN DATEDIFF(YEAR, m.birth_date, GETDATE()) BETWEEN 50 AND 64 THEN '50-64'
    ELSE '65+'
END AS age_group,
c.diagnosis_code,
d.disease_name
FROM claims c
JOIN members m ON c.member_id = m.member_id
JOIN diagnoses d ON c.diagnosis_code = d.diagnosis_code;
```

100 % 99+ 0 Ln: 75,

Results Messages

	member_id	birth_date	age_group	diagnosis_code	disease_name
1	1006	1985-05-24	35-49	C50	Breast Cancer
2	1033	2000-03-13	18-34	E11	Type 2 Diabetes
3	1381	1958-03-13	65+	I25	Coronary Artery Disease
4	1358	1952-10-04	65+	J45	Asthma
5	1458	1965-03-11	50-64	J45	Asthma
6	1223	1955-02-12	65+	I25	Coronary Artery Disease

Query executed successfully. localhost (17.0 RTM) LAPTOP-MFQDC801\Nabeel... PopulationHealthP

SQLQuery2_1...abeel (78))*

```sql
-- Step 6 – Combine Count + Total Members to Get Prevalence

SELECT
c.age_group,
c.disease_name,
c.members_with_disease,
CAST(ROUND(c.members_with_disease * 100.0 / t.total_members, 2) AS DECIMAL(5,2)) AS prevalence_percent
FROM
(
    SELECT
    CASE
        WHEN DATEDIFF(YEAR, m.birth_date, GETDATE()) BETWEEN 0 AND 17 THEN '0-17'
        WHEN DATEDIFF(YEAR, m.birth_date, GETDATE()) BETWEEN 18 AND 34 THEN '18-34'
        WHEN DATEDIFF(YEAR, m.birth_date, GETDATE()) BETWEEN 35 AND 49 THEN '35-49'
        WHEN DATEDIFF(YEAR, m.birth_date, GETDATE()) BETWEEN 50 AND 64 THEN '50-64'
        ELSE '65+'
    END AS age_group,
    d.disease_name,
    COUNT(DISTINCT c.member_id) AS members_with_disease
    FROM claims c
```

100 % 99+ 0 Ln: 180, Ch: 47 (2184 chars, 55 lines) SPC

Results Messages

	age_group	disease_name	members_with_disease	prevalence_percent
1	18-34	Type 2 Diabetes	67	60.91
2	18-34	Asthma	63	57.27
3	18-34	Coronary Artery Disease	63	57.27
4	18-34	Hypertension	62	56.36
5	18-34	Breast Cancer	59	53.64

Query executed successfully. localhost (17.0 RTM) LAPTOP-MFQDC801\Nabeel... PopulationHealthProject 00:00:00 Row: 1, Col:

Eligibility Analysis

SQLQuery2_1...abeel (78))*

```sql
-- Step 1 - Identify Valid vs Invalid Claims

SELECT
c.claim_id,
c.member_id,
m.enrollment_date,
c.service_date,
CASE
    WHEN c.service_date >= m.enrollment_date THEN 'Valid'
    ELSE 'Invalid'
END AS claim_status
FROM claims c
JOIN members m
ON c.member_id = m.member_id
```

100 % 99+ 0

Results Messages

	claim_id	member_id	enrollment_date	service_date	claim_status
1	5001	1006	2023-01-06	2023-12-10	Valid
2	5002	1033	2023-01-02	2024-01-14	Valid
3	5003	1381	2021-02-19	2023-10-17	Valid
4	5004	1358	2021-04-27	2024-01-08	Valid
5	5005	1458	2022-08-11	2023-09-11	Valid
6	5006	1223	2022-06-07	2024-09-11	Valid
7	5007	1237	2021-08-26	2024-08-25	Valid
8	5008	1485	2022-08-16	2023-08-27	Valid
9	5009	1378	2021-08-13	2024-10-28	Valid
10	5010	1141	2023-04-30	2023-09-10	Valid

Query executed successfully. localhost (17.0 RTM) LAPTOP-MFQDC80

```sql
-- Step 2 - Count Invalid Claims

SELECT
COUNT(*) AS total_claims,
SUM(CASE WHEN c.service_date < m.enrollment_date THEN 1 ELSE 0 END) AS invalid_claims,
SUM(CASE WHEN c.service_date >= m.enrollment_date THEN 1 ELSE 0 END) AS valid_claims
FROM claims c
JOIN members m
ON c.member_id = m.member_id
```

00 % 99+ 0 Ln: 201, Ch: 1 (304

Results Messages

total_claims	invalid_claims	valid_claims
2000	94	1906